

EEG Microstates: A Compact View of Topographic Dynamics

Six key points · handout / poster companion

1 Recurring topographies



EEG scalp maps change continuously, but certain spatial configurations recur. Around moments of high global field power (GFP), these dominant topographies can be especially well defined and reproducible.

2 A low-dimensional summary



Microstate analysis clusters similar scalp topographies into a small number of representative classes. It provides a **low-dimensional description of the continuously evolving global EEG field**.

3 Dominant, not exclusive



A microstate class reflects the currently dominant topographic configuration. It does not imply that the whole brain is in a single mutually exclusive physiological state; multiple generators and networks may contribute in parallel.

4 Temporal dynamics become measurable



Discretization turns ongoing topographic fluctuation into a symbolic sequence such as **A → C → B → D**, enabling analysis of duration, occurrence, coverage, transitions, syntax, entropy and higher-order temporal structure.

5 Only part of the EEG state



The class label mainly captures spatial configuration. Other relevant features - amplitude, spectral content, phase, connectivity and the precise mixture of neural generators - are not fully represented by the label alone.

6 A useful coarse-grained representation



Microstates are neither arbitrary labels nor complete physiological states. They are **recurring dominant configurations of the global EEG field** that provide a compact, physiologically meaningful representation of brain dynamics.



Take-home message: Microstates capture dominant recurring configurations of the EEG field - a useful coarse-grained representation of brain dynamics, rather than a complete description of the underlying physiological state.



Interactive visualization

<https://christoph-berger.github.io/microstates/>